

Assignment 1

1. Prove that

(a) $P(A_1 + A_2 + \cdots + A_n) \leq P(A_1) + P(A_2) + \cdots + P(A_n)$ [Boole's inequality]

(b) $P(A_1 A_2 \cdots A_n) \geq \sum_{i=1}^n P(A_i) - (n-1)$ [Bonferroni's inequality]

2. For any three events A, B and C , prove that

(a) $P(\bar{A} + B) = 1 - P(A) + P(AB)$.

(b) $P(A + B \mid C) = P(A \mid C) + P(B \mid C) - P(AB \mid C)$.

(c) $P(A\bar{B} \mid C) + P(AB \mid C) = P(A \mid C)$

(d) $P(X \mid C) = P(A \mid C) + P(B \mid C)$ where $X = A + B$ if $P(ABC) = 0$.

3. (a) A missile was fired at a plane on which there are two targets T_1 and T_2 . The probability of hitting T_1 is p_1 and that of hitting T_2 is p_2 . It is known that T_2 was not hit. Find the probability that T_1 was hit. [Ans. $\frac{p_1}{1-p_2}$]

(b) A die is thrown twice, the event space S consists of the 36 possible pairs of outcome (a, b) , each assigned with probabilities $1/36$. Let A, B, C denote the events

$$A = \{(a, b) \mid a \text{ is odd}\}$$

$$B = \{(a, b) \mid b \text{ is odd}\}$$

$$C = \{(a, b) \mid a + b \text{ is odd}\}$$

Check whether A, B and C are mutually independent or pairwise independent.

4. (a) A problem in statistics is given to three students A, B and C whose chances of solving it are $\frac{1}{2}$, $\frac{3}{4}$ and $\frac{1}{4}$, respectively. What is the probability that the problem is solved? [Ans. $\frac{29}{32}$]

(b) From a vessel containing 3 white and 5 black balls, 4 balls are transferred into an empty vessel. From this vessel a ball is drawn and it is found to be white. What is the probability that out of four balls transferred 3 are white and 1 black. [Ans. $\frac{1}{7}$]

5. (a) In a bolt factory, machines A, B, C manufacture respectively 25%, 35%, 40% of the total. Of their output 5%, 4% and 2% are defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it was manufactured by machine A, B and C respectively? [Ans. $\frac{25}{69}, \frac{28}{69}, \frac{16}{69}$]

(b) If A, B, C are mutually independent events, then prove that A and $B + C$ are independent.

6. (a) Two persons agree to play a game by drawing balls by turn from a box containing 4 white and 6 black balls. He who draws the first white ball wins. Find the probability that the man who starts the game loses the game. [Ans. $\frac{8}{21}$]

(b) In an office 40% employees have scooters and 20% have cars. Among those who have scooters 80% do not have cars. What is the probability that an employee has a car given that he does not have a scooter? [Ans. 0.2]